MERT CAN

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Education

TOBB University of Economics and Technology (TOBB ETU), Ankara, Türkiye **2020 – 2025**
B.Sc. in Computer Engineering, GPA: 3.19 / 4.00
University Entrance Rank: 8,922

Experience

Computer Vision Engineer *SolarVis, Ankara* **Dec 2025**

- Built an end-to-end system to extract 2D roof structures from satellite images and generate 3D roof models from them.
- Used computer vision methods such as image segmentation, depth estimation, and image processing, together with linear algebra–based solutions for 3D reconstruction.
- Gained strong experience in building full AI pipelines, including data labeling, model development, and post-processing using Python, PyTorch, and OpenCV.

AI Compiler Engineer *Synnada AI, Ankara* **May 2025 – Oct 2025**

- Designed and implemented the CUDA backend from scratch for a multi-backend AI compiler, optimizing GPU kernels for performance.
- Improved the C backend by refactoring core operations, enhancing precision and broadcast support, and increasing compiler stability.
- Researched AI compiler design, system-level programming, and GPU architectures, strengthening expertise in C, C++, Python, and CUDA.

Computer Vision Intern *Havelsan, Ankara* **Sep 2024 – Dec 2024**

- Developed object tracking and motion analysis pipelines using computer vision techniques.
- Integrated deep learning–based detection and tracking methods for pedestrian tracking and motion classification.
- Gained deeper understanding of camera geometry, depth estimation, and visual perception principles.

AI Hardware Research Intern *Barcelona Supercomputing Center (BSC), Spain* **Jan 2024 – Apr 2024**

- Developed FPGA-based digital systems and implemented a convolutional neural network (CNN) in Verilog.
- Utilized PS and PL logics on SoC platforms to study FPGA power behavior.
- Explored Large Language Models (LLMs), focusing on the LLaMA family and hardware efficiency aspects.

Processor Architecture Research Intern *Kasırga Microprocessors Lab, Ankara* **Nov 2022 – Nov 2023**

- Conducted one year of research (4 months as an intern).
- Conducted extensive research on processor architecture and hardware design.
- Contributed to the development and enhancement of custom RISC-V–based processor pipelines. Designed and integrated multiple processor units.

Selected Projects

Teknofest RISC-V Processor Design Competition *Nov 2022 – Apr 2023*

- Contributed to the design of a pipelined RV32IMCX processor by implementing the ALU and cache architecture, and assisting in verification and testing stages.
- Achieved 3rd place nationally in the Teknofest RISC-V Design Competition.

Yoga Pose Estimation Using Computer Vision *Mar 2025 – Apr 2025*

- Developed a yoga pose classification system for five core postures using computer vision techniques.
- Applied pose estimation (YOLO, MediaPipe), CNN architectures (EfficientNetB0, MobileNetV2, ResNet18), and classical image processing methods.
- Gained extensive experience in Python, PyTorch, machine learning and computer vision techniques.

Skills

Programming: Python, C, C++, CUDA, Verilog, PyTorch
Specialization: AI Compilers, GPU Programming, Embedded AI, Computer Vision
Tools: Git, Linux

References

- **Sami Can Tandoğdu**
  Co-Founder and CEO in Synnada
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- **Mehmet Ozan Kabak**
  Co-Founder and CEO in Synnada
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